FACIAL EMOTIONAL RECOGNITION USING ARTIFICIAL INTELLIGENCE FOR AUTISTIC CHILDREN

Lara Hamawy[1], Elio Imad[1], Alaa ElHabouche[1], Marwan AlHawat[1], Mohamad Abou Ali[1,2], Ahmad Diab[1]

[1]Department of Biomedical Engineering, Lebanese International University, Beirut, Lebanon
[2] Department of Computer Science & Artificial Intelligence, University of the Basque Country, Spain

2025 Eighth International Conference on Advances in Biomedical Engineering (ICABME) | 979-8-3315-5389-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICABME66883.2025.11211634

Abstract*— Bringing clarity to emotional expression stands at the core of our research. This study focuses on elucidating emotional expression, especially in therapeutic settings for autistic children, where expressing and recognizing emotions can be challenging. Our aim is to explore the relationship between facial expressions, emotional states, and sophisticated machine learning models for real-time recognition. We propose a framework using CNNs, with transfer learning from ResNet50 and VGG16, for both image-based and real-time emotion classification. The system consists of two components: real-time recognition via webcam and classification from curated datasets of autistic children. Our dataset includes emotions such as joy, sadness, fear, anger, surprise, and natural. A number of CNN-based models were put into practice and contrasted. According to the results, the Convolutional Neural Network (CNN) model achieved the highest performance, surpassing both ResNet50 and VGG16. Specifically, the CNN reached an accuracy of 90% on dataset-based (static) images and 88% on real-time images, demonstrating strong consistency in both controlled and dynamic environments. In comparison, ResNet50 achieved 88% on static images and 84% on real-time images, while VGG16 lagged with 78% and 76%, respectively. These findings underscore the effectiveness of deep learning models, particularly CNNs, in delivering accurate and timely emotional recognition. This capability is especially valuable in therapeutic settings for children with autism, where real-time emotional feedback can support social interaction, enhance emotional understanding, and contribute to more personalized and responsive intervention strategies. By accurately identifying facial expressions, CNN-based systems can act as assistive tools that help therapists and caregivers better engage with autistic children and foster emotional development.

Keywords—Facial emotion recognition, Deep learning, Autism, CNN, Transfer learning, AI, Emotion classification.

I. INTRODUCTION

Facial emotion recognition (FER) is crucial for interpreting social cues, but children with autism spectrum disorder (ASD) often struggle to recognize and express emotions, which affects their social interactions. Recent advances in AI and deep learning offer promising tools to support emotional understanding in machine learning [1]. By analyzing facial features, these systems can accurately detect emotions like joy, sadness, fear, anger, surprise, and natural, and provide real-time feedback [2] [3].

The problem statement claims that children with autism spectrum disorder (ASD) frequently struggle with social interaction and emotional expression, which makes it challenging to appropriately gauge emotional states during therapy and day-to-day activities [4]. Current technologies lack the accuracy and flexibility required for real-time, individualized feedback, and existing techniques such as clinician observation and self-reporting are subjective and unreliable. To enable more efficient and customized interventions for kids with autism, a dependable system that employs facial expression detection is required to provide consistent, real-time emotional tracking.

Our study focuses on putting into practice a two-part experimental setup that uses deep learning models for real-time emotion detection through a live webcam stream and emotion categorization using carefully selected face picture datasets of autistic children. Supporting caregivers and therapists in tracking emotional development, facilitating tailored interventions for children with autism, and determining whether the patient is responding and exhibiting the appropriate emotions following the task or exercise are the primary goals.

The rest of the paper is arranged as follows: related work is presented in section II, the methodology is detailed in section III, results and discussion are described in section IV, conclusion and future work are provided in section V.

II. RELATED WORK

Several studies have made significant contributions to the identification and classification of facial emotions based on facial image analysis.

F. M. Talaat et al. (2023) developed a real-time facial emotion recognition system for autistic children using a deep learning framework integrated with IoT and fog computing. It categorized six emotions (anger, fear, joy, natural, sadness, and surprise) from a dataset of 758 training and 72 testing images. The system utilized Enhanced Deep Learning (EDL) with optimized CNN hyperparameters via Genetic Algorithm and U-net segmentation for precise detection, achieving 99.99% accuracy, outperforming other classifiers. The IoT-based framework alerts parents when critical emotions are detected, aiding real-time interventions. Despite its high performance, scalability is limited by the small dataset, which future research aims to expand [5].

N. Alhakbani (2024) developed a facial emotion recognition system to detect engagement in children using CNNs. It utilized two datasets: the LIRIS-CSE dataset for typically developing (TD) children and a Kaggle dataset for autistic children (ASD), focusing on emotions like happiness, sadness, and surprise. Using transfer learning, a CNN model was adapted for ASD, achieving 99% accuracy for TD and 76% for ASD. The model outperformed other classifiers like

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.

SVM and Random Forest, showcasing its ability to capture complex facial patterns. Limitations included smaller ASD datasets and variability in expressions, with future work aimed at expanding datasets, integrating multimodal signals, and enabling real-time detection [6].

I. Ahmad (2024) used a balanced Kaggle dataset of 2,940 facial images to evaluate six pre-trained CNNs for detecting autism spectrum disorder (ASD). Using transfer learning and image preprocessing techniques, ResNet50 achieved the highest accuracy (92%), outperforming other models. The results demonstrate the potential of transfer learning for efficient ASD detection and suggest future applications in early screening and mobile platforms [7].

P. Radocaj, & G. Martinovic (2025) utilized the Facial Emotion Recognition-Autistic Children (FERAC) dataset, which consists of labeled 770 images of children's faces. This study evaluates and compares the performance of convolutional neural networks (CNNs) and transformer-based deep learning. Notably, the Swin Transformer achieved the highest performance, with an accuracy of 0.8000 and an F1-score of 0.7889. Despite the promising results, they suggest multimodal emotion recognition, model interpretability, and personalization for real-world applications in the future [8].

III. MATERIALS AND METHODS

A. Methodology

The proposed experimental setup consists of two parts: the first focuses on autistic image datasets, and the second involves real-time facial emotion recognition.

The first workflow represented in figure 1 shows the methodology for facial emotion recognition based on images of autistic faces using deep learning and using 3 deep learning techniques.

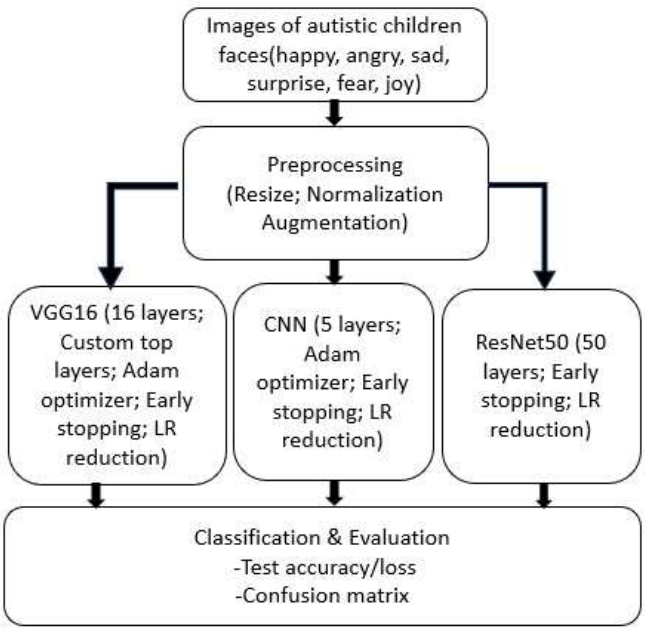

Fig. 1. Workflow based on dataset images using deep learning.

The second workflow represented in figure 2 shows the methodology for facial emotion recognition in real-time using 3 deep learning techniques. Facial emotion recognition (FER) in real time is a technology that detects and classifies human emotions from live video input using machine learning. It involves capturing facial expressions, detecting faces, extracting features, and predicting emotions.

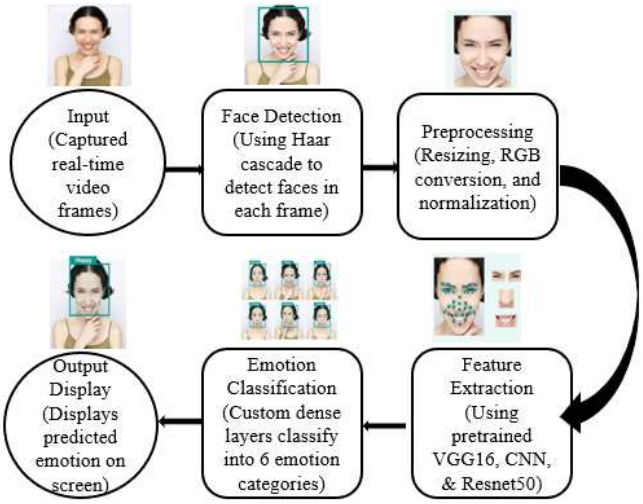

Fig. 2. Workflow based on real-time images using deep learning.

B. Deep learning models and fine tuning

3 network models are selected to be implemented due to their highly effective in processing and analyzing visual data.

Convolutional Neural Networks (CNNs): Used for image tasks like facial emotion recognition. They consist of convolutional layers for feature extraction, activation functions (e.g., ReLU), pooling layers for downsampling, and fully connected layers for classification. A softmax layer at the end outputs the predicted emotion class [9] .

ResNet50: A deep convolutional neural network with 50 layers, known for its use of residual blocks and skip connections**.** These connections help prevent vanishing gradients and allow the model to train effectively at greater depth. ResNet50 is widely used for image tasks like facial emotion recognition due to its strong feature extraction and efficient learning [10].

VGG16: A deep convolutional neural network with 16 layers, mainly using small 3×3 convolution filters and max-pooling layers. It has a simple, uniform structure with ReLU activations after each convolution and ends with fully connected layers for classification. VGG16 is effective for tasks like facial emotion recognition due to its ability to extract detailed image features [10].

Haar cascade: is an algorithm that is capable of detecting objects in images, regardless of their location and scale in an image. Furthermore, this algorithm can run in real-time, making it possible to detect objects in video streams [11].

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.

Adam Optimizer, Learning Rate and Early Stopping: Adam (Adaptive Moment Estimation) optimizer is an adaptive learning rate optimization algorithm adjust learning rates during training. It works well with large datasets and complex models because it uses memory efficiently and adapts the learning rate for each parameter automatically. Early stopping monitors validation loss and terminates training when performance degrades, ensuring better generalization and less overfitting [12].

A comparison is done between these 2 workflows and the methods used.

C. Workflow based on dataset images

i. Dataset

The dataset, named "Autistic Children Emotions", is categorized into six emotional classifications: natural, anger, fear, joy, sadness, and surprise, with each category stored in its folder for systematic access. The images are presented in color, formatted as .jpg files, and uphold a uniform resolution and framing, rendering them appropriate for deep learning applications. In contrast to conventional facial emotion datasets, this particular collection captures the nuanced expressions frequently observed in autistic children, presenting distinct challenges for recognition models. This dataset is prepared by F. Talaat in 2021. [13].

ii. Preprocessing

In the preprocessing pipeline, all images are resized to a fixed resolution of 224×224 pixels to ensure uniform input dimensions for the deep learning model. The images are also converted to RGB format to maintain color consistency across the dataset. Additionally, pixel values are normalized by scaling them to the [0, 1] range, which helps improve the training stability and convergence speed of the model.

iii. Data augmentation

The original dataset showed significant class imbalance. To standardize the dataset, the training data was balanced to a total of 18,000 images, allocating exactly 3,000 images per class across six emotion categories: sadness, anger, joy, surprise, natural, and fear as shown in table I. This was accomplished using data augmentation techniques such as rotation, translation, shear, zoom, horizontal flipping, and brightness adjustments. The process generated between 2,650 and 2,970 synthetic images per class, depending on the initial count. After augmentation, the data was split into 80% training, 10% validation, and 10% test sets, resulting in 14,400 training, 1,800 validation, and 1,800 test images, each evenly distributed across all classes.

TABLE I. DATASET BALANCING AND AUGMENTATION.

Classes	Original Dataset	Balanced & then Augmented Dataset
Fear	30	3000
Natural	48	3000
Surprise	63	3000
Anger	67	3000
Sadness	200	3000
Joy	350	3000

D. Workflow based on real-time images

After training the model, a real-time facial emotion recognition system was implemented using a webcam feed. The system first applies face detection to locate and extract facial regions from each video frame, typically using a Haar Cascade or a pre-trained detector. Once a face is detected, the region of interest is resized to 128×128 pixels, converted to RGB, and normalized to match the input format required by the model. The preprocessed face is then passed to the trained model, which predicts the corresponding emotional state. The predicted emotion label is displayed live on the video stream, allowing for continuous, real-time analysis of facial expressions.

IV. RESULTS & DISCUSSION

This part presents the results of various image-based deep learning models used for emotion classification in autistic children. It includes evaluations of a custom CNN, ResNet, and VGG16 based on the dataset and real-time images.
The table 2 below shows the comparison of emotion recognition model accuracy using dataset and real-time images.

TABLE II. COMPARISON OF EMOTION RECOGNITION MODEL ACCURACY USING DATASET AND REAL-TIME IMAGES.

Model	Based on the dataset images	Based on the real-time images
CNN	90%	88%
ResNet50	88%	84%
VGG16	78%	76%

Since the CNN model achieved the highest accuracy among the tested models, it was selected to demonstrate real-time emotion classification through live webcam predictions. To provide a clear visualization of the model's performance in a real-world scenario, figure 3 below present a series of images captured during live predictions, showcasing the model's ability to detect and classify facial emotions in real-time. The images depict six different emotion classes from the dataset:

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.

sadness, joy, anger, fear, surprise, and **natural**. These examples highlight the model's ability to recognize emotions even with varying lighting conditions, head positions, and facial expressions. The results from these live tests further emphasize the CNN model's robustness and its suitability for deployment in interactive, real-time systems for emotion recognition.

The best model achieved is the CNN for real time images and database images. We will present the accuracy curve for image based-model in figure 4; confusion matrix for image based-model and real-time in model in figures 5 and 6 respectively. Finally, the classification report for image-based model and real-time model is shown in figure 7 and 8 respectively.
The comparison of three deep learning models CNN, ResNet50, and VGG16 reveals varying levels of accuracy when evaluated on both dataset images and real-time images.

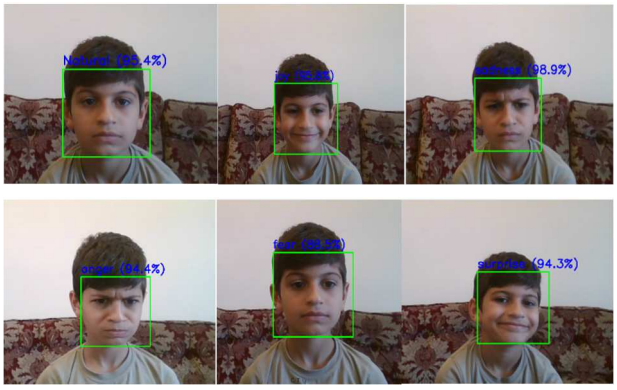

Fig. 3. Detected Facial Emotions with Confidence Scores in Real-Time Using the Best Model CNN.

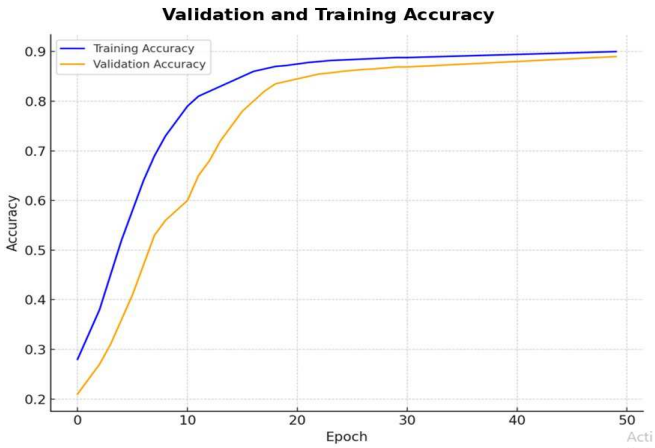

Fig. 4. Training and validation accuracy curve of CNN using the image-based model.

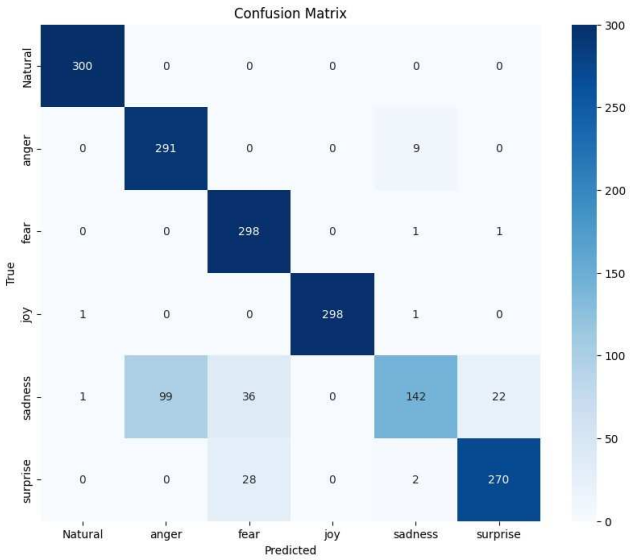

Fig. 5. Confusion matrix of CNN using image-based model.

Fig. 6: Confusion matrix of CNN based on real-time images.

Class	Precision	Recall	F1-Score	Support
Natural	0.78	1.00	0.88	7
Anger	1.00	1.00	1.00	3
Fear	0.60	1.00	0.75	3
Joy	0.95	0.90	0.93	41
Sadness	0.85	0.79	0.82	14
Surprise	0.56	0.83	0.67	6
Accuracy			0.89	74

Fig.7: Classification report of the CNN image-based model.

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.

Class	Precision	Recall	F1-Score	Support
Natural	1.00	1.00	1.00	300
Anger	0.75	0.97	0.85	300
Fear	0.86	0.99	0.92	300
Joy	1.00	0.99	0.99	300
Sadness	0.91	0.47	0.62	300
Surprise	0.92	0.90	0.91	300
Accuracy			0.88	1800

Fig. 8. Classification report of CNN based on real-time.

- CNN shows the highest performance, achieving 90% accuracy on dataset images and 88% on real-time images. Its relatively simple yet effective architecture makes it highly adaptable to the controlled environment of datasets and also generalizes well to real-time conditions. This slight drop indicates minor sensitivity to image variability, but overall robustness.
- ResNet50 follows closely, with 88% on dataset images and 84% on real-time images. The residual connections in ResNet50 help in training deeper networks efficiently. However, the decrease in real-time performance suggests that it might be more sensitive to noise and lighting changes present in real-world scenarios.
- VGG16 ranks lowest, achieving 78% and 76% respectively. Despite its strong hierarchical feature extraction, the large parameter size and lack of skip connections may reduce its generalization, particularly in real-time settings where speed and adaptability are critical.
- For image-based model, we excellent classification accuracy for natural, anger, fear, and joy categories, with very few misclassifications whereas sadness and surprise Sadness and surprise predictions seem to be confused with other emotional states.
- For real-time based model, sadness and surprise categories seem to have higher confusion comparing with other classes.

Overall, CNN offers the best trade-off between accuracy and real-time performance, making it the most practical choice for deployment. ResNet50, while powerful, might require additional preprocessing or fine-tuning to improve real-world robustness. VGG16, although historically significant, shows limitations in adapting to dynamic input conditions.

Comparing our findings with related work, other studies' limitation is in the number of emotions that our approach covers in many classes; also, it takes into account real-time systems based on webcams. In addition, our deep models surpass some studies in accuracy. This leads to an efficient facial emotion recognition for supporting autism therapy.

V. CONCLUSION & FUTURE WORK

Using a labeled dataset of facial expressions, this study explored the development and evaluation of deep learning models for emotion recognition in autistic children. Three models were tested: a custom CNN, ResNet50, and VGG16, evaluated based on test accuracy and real-time webcam performance. Key preprocessing steps included resizing, normalization, data augmentation, and class balancing. The custom CNN outperformed the others with 88% test accuracy and better real-time robustness to lighting and movement. ResNet50 and VGG16 followed with 84% and 76%, respectively. Confusion matrices showed strong detection of distinct emotions like joy and surprise, but confusion persisted between similar emotions, particularly in VGG16. Despite improvements from augmentation and balancing, overfitting and class imbalance remained challenges. The results demonstrate CNNs' capacity for real-time emotion recognition in autism assistance, particularly when applied to small datasets.

Although these findings are encouraging, further research should concentrate on:

Model optimization (such as lightweight models, batch normalization, and dropout), extension of data using fake and varied samples, multimodal methods (physiological signs, voice, and gestures), testing in the real world over time, and personalized systems that adapt to individual emotional patterns.

VI. REFERENCES

[1] W. C. Shiel, 19 July 2016. [Online]. Available: https://www.medicinenet.com/autism_pictures_slideshow/article.htm..

[2] K. Cherney and J. Seladi-Schulman, "What to know about ASD," 24 December 2024. [Online]. Available: https://www.healthline.com/health/autism.

[3] S. Hyman, S. Levy and S. Myers, "Identification,Evaluation and management of children with autism spectrum disorder," *Pediatrics,* p. 145, January 2020.

[4] A. Alharbi, H. Alyami, S. Alenzi, S. Alharbi and Z. Bassfar, "Diagnosing a Child with Autism using Artificial Intelligence," *IJCSNS International Journal of Computer Science and Network Security,* vol. 22, no. 6, p. 145, 2022.

[5] F. M. Talaat, "Real-time facial emotion recognition system among children with autism based on deep learning and IoT.," *Neural Comput & Applic,* no. 35, 7 March 2023.

[6] N. Alhakbani, "Facial Emotion Recognition-based Engagement Detection in Autism Spectrum Disorder," *International Journal of Advanced Computer Science and Applications(IJACSA),* vol. 15, no. 3, 2024.

[7] I. Ahmad, J. Rashid, M. Faheem, A. Akram, N. A. Khan and R. Amin, "Autism spectrum disorder detection using facial images: A performance comparison of pretrained convolutional neural networks," *Healthcare technology letters,* 8 January 2024.

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.

[8] P. Radočaj and G. . Martinović, "Emotion Recognition in Autistic Children Through Facial Expressions Using Advanced Deep Learning Architectures," *Applied Sciences,* vol. 15, no. 17, 2025.

[9] S. Indolia, A. K. Goswami, S. Mishra and P. Asopa, "Conceptual Understanding of Convolutional Neural Network- A Deep Learning Approach," *Procedia Computer Science,* vol. 132, pp. 679-688, 2018.

[10] M. Gori, A. Betti and S. Melacci, in *Machine Learning*, 2nd ed., Chapter 5, 2024, pp. 217-337.

[11] S. Ali, J. Al' Ameri and T. Abbas, "Face Detection Using Haar Cascade Algorithm," in *Fifth College of Science International Conference of Recent Trends in Information Technology (CSCTIT)*, Iraq, 2022.

[12] "Massed Compute," [Online]. Available: https://massedcompute.com/. [Accessed 2024].

[13] F. Talaat, "kaggle," 2023. [Online]. Available: https://www.kaggle.com/datasets/fatmamtalaat/autistic-children-emotions-dr-fatma-m-talaat/data.

Authorized licensed use limited to: Punjab Engineering College (Deemed to be University). Downloaded on March 18,2026 at 09:57:00 UTC from IEEE Xplore. Restrictions apply.